

InternVideo2: Scaling Foundation Models for Multimodal Video Understanding

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<https://github.com/OpenGVLab/InternVideo/tree/main/InternVideo2>

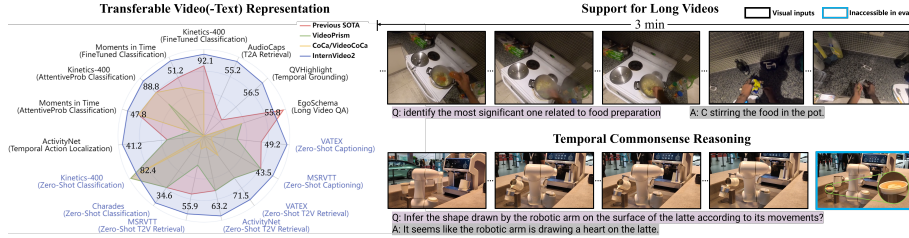


Fig. 1: InternVideo2 yields strong transferable visual and visual-linguistic representations for action recognition, video-text understanding, to video-centric dialogue. It also exhibits capability of long-form video understanding and procedure-aware reasoning.

Abstract. We introduce InternVideo2, a new family of video foundation models (ViFM) that achieve the state-of-the-art results in video recognition, video-text tasks, and video-centric dialogue. Our core design is a progressive training approach that unifies the masked video modeling, crossmodal contrastive learning, and next token prediction, scaling up the video encoder size to 6B parameters. At the data level, we prioritize spatiotemporal consistency by semantically segmenting videos and generating video-audio-speech captions. This improves the alignment between video and text. Through extensive experiments, we validate our designs and demonstrate superior performance on over 60 video and audio tasks. Notably, our model outperforms others on various video-related dialogue and long video understanding benchmarks, highlighting its ability to reason and comprehend longer contexts.

Keywords: Video Foundation Model · Progressive Training · MLLM.

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1 Introduction

Learning transferrable spatiotemporal representations is a critical research area in computer vision, holding diverse applications across domains such as video search [31], game control [7], robotic learning [26], self-driving [103], and scientific studies [84]. Recently, the advancement of Large Language Models (LLMs) [6, 67, 86, 87] and their multimodal variations (MLLMs) [33, 56, 68, 84] have had a profound impact on vision research and other disciplines. Embedding videos effectively into these large models and harnessing their capabilities to enhance video understanding performance has emerged as pivotal tasks [48, 61].

Previous research has identified several effective learning schemes for video representations, including reconstructing videos with masked inputs [28, 37, 85, 90], aligning videos with languages [51, 52, 96, 98], and predicting the next token using videos [1, 49, 83]. These approaches have turned out to be complementary and can be unified through a progressive training scheme. Notably, methods such as InternVideo [94], UMT [51], and VideoPrism [109] have utilized a two-stage training approach involving masked reconstruction and multimodal contrastive learning, leading to improved performance in downstream tasks. Following this line, we aim to further extend this progressive training scheme by incorporating video-based next token prediction and scaling the entire training process, including models and data, to build a new family of video foundation models.

The proposed video foundation model, coined as InternVideo2, is built through a progressive training scheme. The learning involves three stages: (1) capturing spatiotemporal structure via unmasked reconstruction, (2) aligning with semantics from other modalities, and (3) enhancing its open-ended dialogue power through next token prediction. In the initial stage, the model learns to reconstruct the unmasked video tokens, allowing the video encoder to develop basic spatiotemporal perception capability. To estimate the existing tokens, vision encoders (InternViT [17] and VideoMAE-g [90]) trained differently are employed as proxies. In the next stage of crossmodal learning, the architecture is expanded to include audio and text encoders. This not only improves the alignment between videos and text but also endows InternVideo2 the ability to handle video-audio tasks. By incorporating these additional modalities, the model’s understanding of videos is enriched and aligned with their semantics. Finally, in the next-token prediction stage, a video-centric dialogue system and the corresponding instruction-finetuning dataset are built to further tune the InternVideo2. By connecting InternVideo2 to LLMs, the video encoder is further updated through next-token prediction training, enhancing its ability for open-ended tasks such as VQA and video description.

For the training of InternVideo2, we emphasize the spatiotemporal consistency and labeling quality in the data. We build a large-scale multimodal video-centric dataset consisting of 402M data entries, which includes 2M videos, 50M video-text pairs (from WebVid [3] and InternVid [93]), 50M video-audio-speech-text pairs (InternVid2), and 300M image-text pairs. Specifically, for InternVid2, we segment videos into clips semantically and focus on recalibrating the clip descriptions using three modalities: audio, video, and speech. We first generate captions for

these three modalities separately. Then individual captions are fused together to create a more comprehensive description, which will improve the model’s ability to comprehend and interpret the video accurately.

We evaluate InternVideo2 across a wide range of video-related tasks. These tasks span from basic spatiotemporal perception, such as action recognition, to high-level reasoning tasks, such as long video or procedure-aware question-answering (QA), as given in Fig. 1. The results (in Sec. 5) demonstrate that InternVideo2 achieves the state-of-the-art performance on multiple tasks and is able to analyze and reason over sequences of actions. This top performance signifies its capability to effectively capture and understand video content. These empirical findings validate that InternVideo2 could serve as a general video encoder for future exploration in video understanding. In summary, our contributions to video understanding are as follows.

- This paper introduces InternVideo2, a competitive family of video foundation models that leverages masked reconstruction, crossmodal contrastive learning, and next token prediction to make model perceptive, semantic, and capable of reasoning in video understanding.
- InternVideo2 achieves the state-of-the-art performance for more than 60 video / audio tasks. Our model demonstrates superior performance in video-related dialogue and long video understanding, highlighting its potential in modeling high-level world knowledge.
- We provide an enhanced dataset to train InternVideo2. This includes the validation and incorporation of audio data during training, as well as the improved captioning method. These improvements result in significant enhancements in model performance and generalization ability.

2 Related Work

Video Foundation Models. Studies on learning video foundation models becomes increasingly crucial considering its wide applications [?, 51, 52, 85, 89, 90, 92, 94, 96, 98, 109]. Typical methods in building video foundation models (ViFM) include video-text contrastive learning [52, 94, 96], masked video modeling [30, 85, 90, 94], and next token prediction [1, 80, 83]. Specifically, All-in-one [89] utilized a single backbone with unified multiple pretraining objectives. On the other hand, UMT [51] combined masked modeling with video-text contrastive learning, demonstrating strong performance in both action recognition and video-language tasks. Another approach is mPLUG-2 [95], which introduced a new design for modulating different modalities. It shared a common module across modalities to enhance relations while incorporating modality-specific modules for discrimination. In addition to video-text pretraining, researchers have also explored the use of audio information in videos to improve performance. MERLOT Reserve [104] learned video representations using a large-scale dataset of video-speech-transcript pairs. VALOR [14] employed independent encoders for video, audio, and text and trains a joint visual-audio-text representation. VAST [15] constructed an audio-visual-speech dataset and develops a multimodal backbone

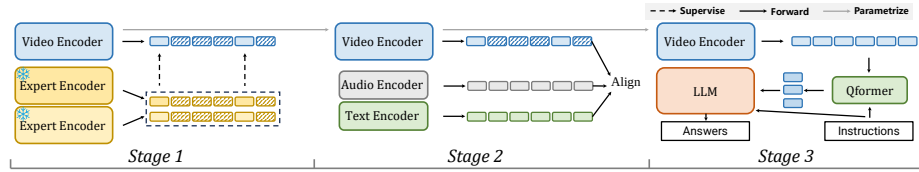


Fig. 2: Framework of InternVideo2. It consists of three consecutive training phases: unmasked video token reconstruction, multimodal contrastive learning, and next token prediction. In stage 1, the video encoder is trained from scratch, while in stages 2 and 3, it is initialized from the version used in the previous stage.

that excels in video-audio-related tasks. VideoPrism [109] combined video-text contrastive learning and video token reconstruction on a combination of public and proprietary videos, achieving leading results across various video tasks.

Multimodal Large Language Models. With advances in large language models (LLMs) [6, 24, 76], their multimodal versions (MLLMs) is becoming popular as it can handle open-world tasks. Seminal works like Flamingo [1] showed outstanding zero/few-shots performances over a range of multimodal tasks [36, 63, 72, 97]. Public MLLMs [33, 56, 113] such as LLaVA [56] and InstructBLIP [21] proposed to use visual instruction-tuning data to improve the visual dialogue ability. Some video-centric MLLMs have been proposed, such as VideoChat [48], VideoChatGPT [61] and Valley [60], by using instruction data to connect video encoders to LLMs for open-world video understanding.

3 Method

We learn InternVideo2 in three stages, illustrated in Fig. 2. The stages include spatiotemporal token reconstruction, video-audio-speech-language contrastive learning, and connecting to a large language model (LLM) for joint training.

Video Encoder. The video encoder used in InternVideo2 follows the Vision Transformer (ViT) [25] and includes additional projection layers for distillation. Inspired by previous works [17, 102], we introduce attention pooling to the ViT. For input videos, we sparsely sample 8 frames [91] and perform a 14×14 ($h \times w$) spatial downsampling. These spatiotemporal tokens are then concatenated with a class token and combined with 3D position embeddings. The hyperparameters used for ViT-1B and -6B are the same as CoCa-1B [102] and InternViT-6B [17], respectively. The details of ViT-6B architecture are given in Supp.

3.1 Stage1: Reconstructing Unmasked Video Tokens

We exploit two expert models to guide the video encoder to conduct token-level reconstruction for unmasked areas. Specifically, we adopt InternVL-6B [17] and VideoMAEv2-g [90] to transfer unmasked knowledge via simple projection layers.

When training, we input the full videos into different teachers and mask out 80% of the tokens frame by frame, under the semantic guidance of the multimodal model InternVL and motion-aware model VideoMAEv2. We only align the unmasked tokens, by minimizing their mean squared error (MSE) between student and teachers. The learning objective is to reconstruct the remaining tokens as:

$$\mathcal{L} = \frac{1}{Z} \sum (\alpha_1 |f^V(\mathbf{V}_p) - h(\mathbf{V}_p)|^2 + \alpha_2 |f^V(\mathbf{V}_p) - g(\mathbf{V}_p)|^2), \quad (1)$$

where f^V , h , and g are our video encoder, InternViT-6B [17], and ViT-g of VideoMAEv2, respectively. p stands for the token index and $f(\mathbf{V}_p)$ is the corresponding token extracted by InternVideo2 for input video $\mathbf{V}$. Z is the normalization factor. α_1 and α_2 balance the influence between the employed models.

In implementation, we align the last 6 layers of InternVL, last 4 layers of VideoMAEv2, and the final output token of InternVL. Different losses are simply summed for optimization. After pretraining, we drop those projection layers and only use the basic encoder. Compared with only using the multimodal model in UMT and VideoPrism, our strategy makes the vision encoder multimodal-friendly as well as enhances its temporal sensitivity for action modeling.

3.2 Stage 2: Aligning Video to Audio-Speech-Text

We exploit the correspondence between video and audio, speech, and text to encourage InternVideo2 to learn more semantics. In practice, InternVideo2 has a huge video encoder, and its employed audio and text encoders are relatively lightweight. The used audio encoder is a 12-layer transformer initialized with BEATs [13] (90M). It takes in audio features, which are 64-dimensional log Mel filterbank spectrograms using a 25ms Hamming window, transformed from 10-second-long clips, padding with zeros. For the text and speech encoders, we initialize the text encoder and multimodal decoder using BERT-Large [24]. Specifically, we utilize the initial 19 layers of BERT-Large as the text encoder, with the subsequent 5 layers equipped with cross-attention layers serving as the multimodal decoder.

For pretraining objectives, we establish alignment across different modalities via text, including video, audio, image, and speech. We employ crossmodal contrastive and matching losses with masked language reconstruction loss as:

$$\mathcal{L} = \mathcal{L}_{\text{CON}} + \mathcal{L}_{\text{MAC}} + \mathcal{L}_{\text{MLM}}, \quad (2)$$

The employed $\mathcal{L}_{\text{MAC}}$ and $\mathcal{L}_{\text{MLM}}$ are standard loss from [19]. Specifically, the crossmodal contrastive learning is given as:

$$\mathcal{L}_{\text{CON}} = \sum_{M, T, M'} \mathcal{L}_{\text{CON}}(M, T, M') = - \sum_{M, T, M'} \left(\sum_{i=1}^N \log \frac{\exp(\text{sim}(f_i^M, f_i^{T, M'})/\tau)}{\sum_{j=1}^N \exp(\text{sim}(f_i^M, f_j^{T, M'})/\tau)} + \sum_{i=1}^N \log \frac{\exp(\text{sim}(f_i^{T, M'}, f_i^M)/\tau)}{\sum_{j=1}^N \exp(\text{sim}(f_i^{T, M'}, f_j^M)/\tau)} \right), \quad (3)$$

where f^V and f^T denote the learned video and text embeddings, respectively.

M and $T_{M'}$ indicates the modality of input signals and the text descriptions describing, respectively. $\text{sim}(\cdot)$ computes the cosine similarity between two features. τ is the learnable temperature.

For the matching part, it is given as:

$$\mathcal{L}_{\text{MAC}} = -y \log f_p(\mathbf{V}, \mathbf{T}) - (1 - y) \log(1 - f_p(\mathbf{V}, \mathbf{T})), \quad (4)$$

where $f_p(\mathbf{V}, \mathbf{T})$ computes the matching likelihood between $\mathbf{V}$ and $\mathbf{T}$. y denotes whether the given video and text are paired ($y = 1$) or not ($y = 0$).

The employed masked language modeling loss is:

$$\mathcal{L}_{\text{MLM}} = -\log f_p^T(\mathbf{T}_j | \mathbf{T}_{<j}), \quad (5)$$

where $f_p^T(\mathbf{T}_j | \mathbf{T}_{<j})$ computes the likelihood of the j_{th} text token based on the previous ones. Here $\mathbf{T}$ refers to video captions.

To improve the training efficiency, we employ the masked learning strategy, aligning unmasked video tokens to tokens from other modalities first, then using full video tokens reconstruction shortly. Specifically, it consists of two steps as follows:

Aligning Masked Visual-Language-Audio. We freeze the audio encoder and focus on aligning visual, audio, and text features. For pre-training, we use a comprehensive set of image, video, and audio-video data. The combinations of modalities used are represented as $\{M, T_{M'}\} \in \{\{I, T_I\}, \{V, T_V\}, \{V, T_{VAS}\}, \{VA, T_{VAS}\}\}$ where each pair denotes the concatenated features from the respective modalities.

Unmasked Visual-Audio-Language Post-Pretraining. We freeze the vision encoder to jointly align audio, visual, and text features. Post-pretraining is conducted using a smaller subset of image and video data (25M samples), along with the full set of audio (0.5M samples) and audio-video data (50M samples). Since the parameters of the largest ViT-6B model are frozen, we do not use masking strategies in this phase to ensure consistency with the inference process and to minimize any performance degradation in downstream tasks. The modality combinations used here are $\{M, T_{M'}\} \in \{\{I, T_I\}, \{V, T_V\}, \{A, T_A\}, \{V, T_{VAS}\}, \{VA, T_{VA}\}\}$.

3.3 Stage3: Predicting Next Token with Video-Centric Inputs

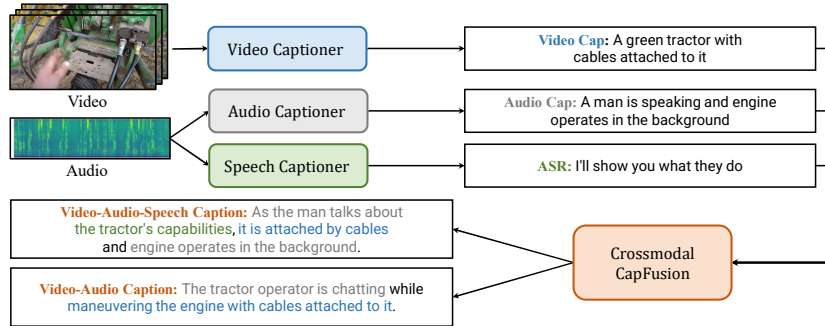
To further enrich the semantics embedded in InternVideo2 and improve its support for video-centric dialogue, we tune it by connecting it to a LLM with QFormer design [46, 47]. We employ the progressive learning scheme in [49] by using InternVideo2 as the video encoder and train a video blip for communicating with open-sourced LLM [40, 111]. In training, the video encoder will be updated. Detailed training receipts are the same with VideoChat2 [49].

4 Multimodal Video Data

We list our training data in Tab. 1. Among the datasets used, KMash and InternVid2 are newly built, whereas the rest are public available.

Table 1: Summary of datasets used in InternVideo2 pretraining process.

Pretraining Stage	Dataset	Domain	# of clips	Annotation
Stage 1	KMash	Web Video	2M	-
Stage 2 (img-txt)	LAION, etc	Web Image	300M	Alt-text / Generated Caps
	WebVid2M	Web Video	250k	Alt-text
	WebVid10M	Web Video	9.7M	Alt-text
	InternVid	Youtube Video	40M	Generated Caption
Stage 2 (vid-txt)	InternVid2	Youtube Video	50M	Generated Caption
	LLaVA, etc	Web Image/Video	2.1M	Conversation, QA

**Fig. 3:** The framework of our video multimodal annotation system, called VidCap, consists of four main components: video, audio, and speech captioners, along with a LLM for integrating captions from these modalities.

4.1 Video-only Data for Masked Autoencoders

We curate a new video set without labels named *K-Mash* from action recognition datasets [10, 35, 38, 65, 108], as detailed in Supp. It encompasses a wide range of video types, including first- and third-person perspectives, with both short and long duration, and featuring various settings. Further, we give $K\text{-Mash}_{2M}$ with additionally sourced and selected 844K videos from YouTube for diversity.

4.2 Videos with Audio-Speech Modalities

We build a multimodal video dataset, coined as InternVid2, with video-audio-speech information and their descriptions for strengthening video perception via other modalities. It consists of 100M videos along with their VAS captions. In InternVid2, we collect videos from several sources (detailed in the supp), segment them into clips, and automatically annotate them based on their unimodal or crossmodal inputs. We highlight the importance of temporal segmentation in clip generation and our video multimodal annotation system. We find refining them leads to notable downstream improvements.

Temporal Consistency Matters. We employ a temporal boundary detection model AutoShot [114] to segment videos into clips instead of SceneDet filter from

FFmpeg. It predicts boundaries based on temporal semantic variations rather than pixel differences, and is able to generate semantically complete cuts without mixing extra frames with inconsistent context.

Video Multimodal Annotation. We design a video multimodal annotation system VidCap to give proper unimodal and crossmodal descriptions for textualizing videos from different perceptions. It automatically captions visual, audio, and speech of InternVid2, then it corrects them and fuses them for cross-modal captions via LLM. The system frame is given in Fig. 3. VidCap has independent video, audio, speech captioner, and a LLM for caption refinement and fusion. For video, speech, and caption postprocessing, we employ existing methods as the video captioning pipeline in [93], WhisperV2-large model [75], and Vicuna-1.5 [111]. For audio, we craft a audio captioner upon VideoChat [48], as we find no open-sourced one. It extracts audio features from inputs by Beats [13]. We learn it by only tuning its Qformer using a combination of the large-scale audio-text corpus WavCaps [64] dataset. Details are given in the supplementary material.

5 Experiments

In our evaluation of InternVideo2, we assess the models from three learning stages. The evaluation covers a wide range of tasks, including video recognition, video retrieval, question-answering, and more. It includes various scenarios such as zero-shot learning, finetuning, and linear probing.

For InternVideo2 trained in stage 1, 2, and 3, we denote them with InternVideo2_{s1}, InternVideo2_{s2}, and InternVideo2_{s3}, respectively. We also learn a CLIP-style InternVideo2 indicated by InternVideo2_{clip}. It is post-pretrained from InternVideo2_{s2} by only preserving video and text encoders and contrastive loss.

Each training stage of InternVideo2-6B uses different configurations and resources. In the first stage, we employ 256 NVIDIA A100 GPUs and train the model for 18 days. The second stage also utilizes 256 A100 GPUs and spans a training period of 14 days. Finally, in the third stage, we use 64 A100 GPUs and train the model for 3 days. We introduce DeepSpeed and FlashAttention [22] for training and inference. More implementation details, experiment results, and ablations are given in the supp.

5.1 Video Classification

Action Recognition. We test InternVideo2 on Kinetics (*i.e.*, K400, 600 and 700 [8–10]), Moments in Time V1 (MiT) [65], Something-Something V2 (SSv2) [35], UCF [79], HMDB [43], Charades [32], ActivityNet [38] (ANet) and HACS [108]. We evaluate in four settings: (a) *end-to-end finetuning* the whole backbone; (b) *attentive probing* is similar to linear pooling, but extra trains the attention pooling layer [102]. (c) *linear probing* which freezes the backbone and only trains the task head; and (d) *zero-shot*.

Table 2: End-to-end finetuning action recognition results (top-1 accuracy) on Kinetics, SomethingSomething, and Moments in Time. [†] denotes that the result is achieved with different resolutions or frame rates.

Method	Training Data	Setting	K400	K600	K700	Sth-Sthv2	MiT	ANet	HACS
CoVeR [106]	IV -3B	16 × 448	87.1	87.9	79.8	70.8	46.1	-	-
Hiera-H [78]	V -0.25M	16 × 224	87.8	88.8	81.1	76.5	-	-	-
CoCa-g [102]	I -3B	16 × 576	88.9	89.4	82.7	-	49.0	-	-
MTV-H [90]	IV -370M	32 × 280	89.9	90.3	83.4	-	-	-	-
VideoMAEv2-g [90]	V -1.35M	64 × 266	90.0	89.9	-	77.0 [†]	-	-	-
V-JEPA-H [4]	V -2M	16 × 224	-	-	-	77.0	-	-	-
MVD-H [92]	IV -1.25M	16 × 224	-	-	-	77.3	-	-	-
UniFormerV2-L [50]	IV -401M	64 × 336	90.0	90.1	82.7	73.0 [†]	47.8 [†]	94.7	95.4
InternVideo [94]	V -12M	<i>ensemble</i>	91.1	91.3	84.0	77.2	-	-	-
InternVideo2 _{s1} -1B	IV -1.1M	8 × 224	91.3	91.4	85.0	77.1	50.8	-	-
InternVideo2 _{s1} -1B	IV -1.1M	16 × 224	91.6	91.6	85.4	77.1	50.9	-	-
InternVideo2 _{s1} -6B	IV -2M	8 × 224	91.9	91.7	85.7	77.5	51.0	-	-
InternVideo2 _{s1} -6B	IV -2M	16 × 224	92.1	91.9	85.9	77.4	51.2	95.9	97.0

Table 3: Attentive probing recognition results (top-1 accuracy) on Kinetics-400/600/700, Moments in Time and Something-Something V2.

Method	Training Data	Setting	K400	K600	K700	MiT	SSV2
UMT-L [51]	IV -25M	-	82.8	-	-	40.3	54.5
VideoMAEv2-g [90]	V -1.35M	-	82.1	-	-	35.0	56.1
V-JEPA-H [4]	V -2M	16 × 384	81.9	-	-	-	72.2
DINOv2-g [69]	I -142M	16 × 224	83.4	-	-	-	50.0
VideoPrism-g [109]	V -619M	16 × 288	87.2	-	-	45.5	68.5
ViT-e [23]	I -4B	128 × 224	86.5	-	-	43.6	-
ViT-22B [23]	I -4B	128 × 224	88.0	-	-	44.9	-
CoCa-g [102]	I -3B	16 × 576	88.0	88.5	81.1	47.4	-
InternVideo2 _{s2} -1B	IV -25.5M	16 × 224	87.9	88.0	79.5	46.3	67.3
InternVideo2 _{s2} -6B	IV -400M	16 × 224	88.8	89.1	81.0	47.8	67.7

End-to-end Finetuning. Tab. 2 shows InternVideo2-6B obtains new state-of-the-art (SOTA) results on Kinetics (**92.1%/91.9%/85.9%** on K400/600/700, respectively), SthSthv2, MiT, ANet, and HACS with only 16 frames, while the previous SOTAs require larger resolution (224 *vs.* 576) or model ensemble. As for MiT in Table 2, InternVideo2-6B exceeds the previous SOTA, CoCa-g, by a significant margin of 2.2% (**51.2%** *vs.* 49.0%). Regarding the temporal-related actions in Table 2, our InternVideo2-6B also surpasses MVD [92] on SSV2 (**77.5%** *vs.* 77.3%). Moreover, our InternVideo2-6B showcases top performance on untrimmed video analysis, as indicated in Table 2, with 95.9% on ActivityNet and 97.0% on HACS. These results affirm our model’s superior capability for robustly identifying complex actions across varied scenes. Note “I” and “V” denotes images and videos, respectively. “IV-3B” means the total number of the used images and videos is 3B, while “I-3B” means using 3B images.

Attentive Probing. As in Tab. 3, InternVideo2-6B not only outperforms ViT-22B [23] and CoCa-g [102] in scene-focused datasets but also surpasses or matches the performance of the latest video foundation model [4,109], on datasets emphasizing temporal dynamics (SthSthV2). This underscores our model’s exceptional ability to understand and interpret both spatial and temporal information effectively.

Table 4: Linear probing action recognition results (top-1 accuracy) on Kinetics-400, Something-Something V2, UCF-101 and HMDB-51.

Method	Setting	K400	SSV2	UCF-101	HMDB-51
VideoMAEv2-H [90]	12×224	25.8	-	56.4	34.1
TVTSv2-H [105]	12×224	73.1	-	91.8	65.7
OpenCLIP-G [20]	8×224	78.3	35.8	90.7	-
DINOv2-g [69]	8×224	78.4	38.3	91.2	-
InternVideo2 _{s1} -1B	16×224	81.6	46.3	96.0	71.6
InternVideo2 _{s1} -6B	16×224	82.0	47.8	96.3	71.8
InternVideo2 _{s2} -6B	16×224	84.2	56.7	97.3	80.7

Table 5: Zero-shot action recognition on UCF, HMDB, MiTv1, SSv2-MC, and Charades.

Method	#F	Training Data	UCF	HMDB	MiT	SSv2-MC	Charades
CLIP [74]	12	I -400M	68.9	43.2	-	29.6	-
TVTSv2 [105]	12	V -8.5M	78.0	52.1	-	48.4	-
VideoCoCa-g [98]	16	V -145M	-	-	-	-	25.8
VideoPrism-g [109]	16	V -619M	-	-	-	-	32.4
InternVideo2 _{clip} -1B	8	IV -25.5M	88.8	53.9	31.6	61.5	32.9
InternVideo2 _{clip} -6B	8	IV -400M	89.5	56.7	32.9	63.5	34.6

Table 6: Zero-shot action recognition results on Kinetics.

Method	#F	K400		K600		K700	
		top-1	AVG	top-1	AVG	top-1	AVG
CLIP [74]	8	58.4	70.1	55.1	67.2	46.1	58.4
EVA-CLIP-L [81]	1	-	65.0	-	64.9	-	59.1
EVA-CLIP-E [81]	1	-	69.8	-	69.3	-	63.4
ViCLIP-L [93]	8	64.8	75.7	62.2	73.5	54.3	66.4
VideoCoCa-g [98]	16	72.0	81.3	-	-	-	-
InternVL-6B [17]	8	69.1	79.4	68.9	78.8	60.6	71.5
EVA-CLIP-18B [82]	16	-	79.4	-	79.4	-	72.2
VideoPrism-g [109]	16	76.4	85.4	-	-	-	-
InternVideo2 _{clip} -1B	8	<u>73.1</u>	<u>82.4</u>	72.8	81.8	64.9	75.2
InternVideo2 _{clip} -6B	8	72.7	82.2	71.7	81.2	64.2	75.2

Linear Probing. In Tab. 4, InternVideo2-1B significantly outperforms the previous SOTA, DINOv2-g [69], by notable margins: +3.2% on K400, +8.0% on SthSthV2, and +4.8% on UCF-101. As we scale the model, an upward trend in results is observed, underscoring the benefits of model enhancement. Notably, the integration of multimodal pretraining (stage 2) yields further rise in results. We suppose stage 2 enhances feature discrimination.

Zero-shot. Table 5 and 6 show InternVideo2 gets 72.7% / 71.7% / 64.2% on K400/600/700, respectively, outperforming others but VideoPrism on K400 (76.4%). On UCF [79], HMDB [43], MiT [65], SSv2-MC, and Charades, InternVideo2 gives an cutting edges over others. The clear gap between VideoPrism and InternVideo2 on K400 may signify the importance of pretraining corpus in VideoPrism (311M videos with text and 36.1M of them are manually labeled) for

Table 7: Finetuned temporal action localization results on THUMOS14 [39], ActivityNet [42], HACS Segment [108] and FineAction [57]. We report average mAP. “Flow” denotes the ensembling I3D flow feature. * denotes the result is achieved with Flow.

Backbone	THUMOS14	HACS	ActivityNet	FineAction
I3D [10] + Flow	66.8	-	35.6	-
R(2+1)D [88]	55.6	-	36.6	-
InternVideo	71.6*	41.3	39.0	17.6
VideoMAEv2-g [90]	69.5	-	-	18.2
InternVideo2 _{s1} -1B	69.8	42.4	40.4	27.2
InternVideo2 _{s1} -6B	72.0	43.3	41.2	27.7

Table 8: Video instance segmentation performance (mAP) on YouTube-VIS19 [99].

Method	Backbone	#Params	YouTubeVIS19
Mask2Former	Swin-L (image) [58]	219M	60.3
Mask2Former	InternViT (image)	6B	63.4
Mask2Former	InternVideo2 _{s1}	6B	64.2

K400 in zero-shot. Note that on the datasets of Kinetics, UCF101 and HMDB51, which have a distribution closer to the pre-training dataset used in stage1, the performance of Internvideo2-6B is slightly inferior to that of Internvideo2-1B. We suppose this is caused by Internvideo2-6B uses a more abundant pretraining dataset in stage2, leading to the forgetting of pretraining data in stage1.

Temporal Action Localization. We evaluate models on four temporal action localization (TAL) datasets: THUMOS14 [39], ActivityNet [42], HACS Segment [108] and FineAction [57] in a feature-based manner with finetuning. We employ output of the 7-th layer from InternVideo2 for inputs as the corresponding features without fusing anything else. ActionFormer [107] is used as the detection head. We report mean Average Precision(mAP) under multiple tIoU as in [12, 54, 100, 107]. In Table 7, InternVideo2-6B gets the highest mAP among all comparisons in all datasets, while InternVideo2-1B nearly surpass other methods except in THUMOS14. We find InternVideo2-6B almost consistently improves mAP with a notable margin from InternVideo2-1B except in FineAction. We suppose scaling model capacity without data refinement cannot nontrivially improve fine-grained discrimination abilities of models. Scaling detailed annotations in training may address this issue.

Video Instance Segmentation. We evaluate on the Video Instance Segmentation (VIS) dataset Youtube-VIS 2019 [99]. Built upon Mask2Former [18], we employ the video encoder of InternVideo2 as backbone with ViT-adapter [16] for features. We also try InternViT [17] for comparisons. In Table 8, InternVideo2 gets the highest mAP among all. This validates its effectiveness in relatively fine-grained spatiotemporal perception.

Table 9: Results of zero-shot video retrieval in both text-to-video (T2V) and video-to-text (V2T) on MSR-VTT, LSMDC, DiDeMo, MSVD, ANet, and VATEX.

Method	MSR-VTT		LSMDC		DiDeMo		MSVD		ANet		VATEX	
	T2V	V2T	T2V	V2T	T2V	V2T	T2V	V2T	T2V	V2T	T2V	V2T
CLIP [74]	30.4	24.2	13.9	11.9	12.7	18.7	40.5	57.2	9.1	13.2	-	-
CLIP4Clip [59]	32.0	-	15.1	-	-	-	38.5	-	-	-	-	-
ViCLIP [93]	42.4	41.3	20.1	16.9	18.4	27.9	49.1	75.1	15.1	24.0	-	-
InternVideo-L [94]	40.7	39.6	17.6	13.2	31.5	33.5	43.4	67.6	30.7	31.4	49.5	69.5
UMT-L [51]	40.7	37.1	24.9	21.9	48.6	49.9	49.0	74.5	41.9	39.4	-	-
VideoCoCa-g [98]	34.4	64.7	-	-	-	-	-	-	34.5	33.0	53.2	73.6
VideoPrism-g [109]	39.7	71.0	-	-	-	-	-	-	52.7	50.3	62.5	77.1
InternVideo2 _{s2} -1B	51.9	50.9	32.0	27.3	57.0	54.3	58.1	83.3	60.4	54.8	70.4	85.4
InternVideo2 _{s2} -6B	55.9	53.7	33.8	30.1	57.9	57.1	59.3	83.1	63.2	56.5	71.5	85.3

Table 10: Results of finetuning video retrieval in both text-to-video (T2V) and video-to-text (V2T) on MSR-VTT, LSMDC, DiDeMo, MSVD, ANet, and VATEX.

Method	MSR-VTT		LSMDC		DiDeMo		MSVD		ANet		VATEX	
	T2V	V2T	T2V	V2T	T2V	V2T	T2V	V2T	T2V	V2T	T2V	V2T
CLIP [74]	38.2	38.7	22.5	22.6	32.2	33.9	-	-	26.1	26.9	-	-
CLIP4Clip [59]	45.6	45.9	24.3	23.8	43.0	43.6	45.2	48.4	40.3	41.6	-	-
ViCLIP [93]	52.5	51.8	33.0	32.5	49.4	50.2	-	-	49.8	48.1	-	-
UMT-L [51]	58.8	58.6	43.0	41.4	70.4	65.7	58.2	82.4	66.8	64.4	72.0	86.0
InternVideo2 _{s2} -6B	62.8	60.2	46.4	46.7	74.2	71.9	61.4	85.2	74.1	69.7	75.5	89.3

5.2 Video-Audio-Language Tasks

We evaluate InternVideo2 on video retrieval, captioning, and multi-choice question-answering (QA). The former two tasks are conducted by matching video representation and the candidate text ones using the text encoder in stage 2. The latter is tested by the VideoLLM learned in stage 3. We also test audio tasks.

Video Retrieval. We evaluate the video retrieval on six popular benchmarks [2, 11, 38, 77, 97], as shown in Tab. 9 and 10. In eval, eight frames from the input videos are uniformly sampled. *We report $R@1$ scores for both text-to-video (t2v) and video-to-text (v2t) tasks* in Tab. 9 and 10. $R@5$ and $R@10$ are given in Supp.

Table 9 and 10 demonstrate that InternVideo2 outperforms other state-of-the-arts with a notable margin in both t2v and v2t of all used datasets no matter in zero-shot or finetuned settings, except for the v2t of MSR-VTT, where VideoPrism gives the best result. This shows the video-language semantic alignment of transferrity of InternVideo2.

Video Temporal Grounding. We evaluate InternVideo2 on two video temporal grounding (VTG) datasets: Qvhighlight [44], and Charade-STA [32]. The eval setting and used features are the same as in TAL. We use CG-DETR [66] as the grounding head. We report $R1@0.3$, $R1@0.5$, $R1@0.7$, and mAP for moment retrieval as in [44, 53, 66]. Highlight Detection is evaluated in terms of “Very Good” mAP and HiT@1. In Table 11, InternVideo2-1B and InternVideo2-6B bring

Table 11: Finetuned temporal grounding on QVHighlight [44] and Charade-STA [32].

Feature	R1@0.5	R1@0.7	mAP	mAP	HiT@1	Feature	R1@0.3	R1@0.5	R1@0.7	mIoU
CLIP [74]	64.97	48.65	42.96	39.83	64.19	CLIP [74]	65.62	52.77	30.16	45.85
CLIP+SlowFast [29]	65.43	48.38	42.86	40.33	66.21	CLIP+SlowFast [29]	70.43	58.44	36.34	50.13
InternVideo2 _{s2} -1B	<u>70.00</u>	<u>54.45</u>	<u>47.02</u>	<u>42.36</u>	<u>69.74</u>	InternVideo2 _{s2} -1B	<u>78.41</u>	<u>68.36</u>	<u>45.03</u>	<u>57.12</u>
InternVideo2 _{s2} -6B	71.42	56.45	49.24	42.90	72.00	InternVideo2 _{s2} -6B	79.70	70.03	48.95	58.79

(a) QVHighlight

(b) Charade-STA

Table 12: Audio retrieval results on AudioCaps [41], Clothov1, and Clothov2 [27]. We report text-to-audio R@1 accuracy in zero-shot and finetuning settings.

Method	Zero-shot			Finetuning		
	AudioCaps	ClothoV1	ClothoV2	AudioCaps	ClothoV1	ClothoV2
VIP-ANT [110]	27.7	-	-	-	-	-
VAST [15]	-	-	-	52.0	25.1	26.9
LanguageBind [112]	-	12.1	12.1	-	-	-
InternVideo2 _{s2} -6B	37.1	17.4	17.4	55.2	25.3	27.2

Table 13: Results of AudioQA on ClothoQA [55] and Audio-MusicAVQA (AMAVQA) [5], and audio classification on the ESC-50 [71]. Both in the finetuning setting.

Backbone	ClothQA	AMAVQA	Method	Top-1 Acc
AquaNet [55]	14.78	65.59	AST [34]	95.60
MWAFM [45]	22.24	67.54	BEATs [13]	98.10
InternVideo2 _{s2}	30.14	80.51	InternVideo2 _{s2}	98.60

(a) ClothoQA and AMAVQA

(b) ESC-50

gradual performance improvements compared to CLIP [73] and CLIP [73]+Slow-fast [29]. This suggests that a larger spatiotemporal model is more beneficial to short-term video semantic alignment capabilities.

Audio-related Tasks. We evaluate InternVideo2’s audio and text encoders on audio tasks, including audio-text retrieval on AudioCaps [41], Clothov1, and Clothov2 [27]; audioQA on ClothoQA [55] and Audio-MusicAVQA [5]; and audio classification on the ESC-50 [71]. As shown in Tab. 12, 13a, and 13b, our model achieves state-of-the-art performance on all downstream tasks. Considering the limited size of the used audio and text encoders, these audio-related results show crossmodal contrastive learning’s benefits are mutual to the used modalities. Audio and the corresponding text models also gain from this learning.

5.3 Video-centric Dialogue and its Applications

Tab. 14 shows results on MVBench [49], Egoscema [62], and Perception Test [70], by equipping VideoChat2 [49] with our InternVideo2 encoder. VideoChat2-InternVideo2 demonstrates a significant performance advantage over other systems

Table 14: Results of Chat-centric Evaluation on Multiple Choice Video-QA on MVBench [49], Egoschema and Perception Test.

Model	ViEncoder	LLM	MVBench	Egoschema	Perception Test
GPT-4V [68]	-	GPT-4	43.5	-	-
Gemini 1.0 Pro [84]	-	-	37.7	55.7	51.1
Gemini 1.0 Ultra [84]	-	-	-	61.5	54.7
Gemini 1.5 Pro [84]	-	-	-	72.2	-
LLaVA-Next-Video	CLIP-L	Vicuna-7B	46.5	43.9	48.8
VideoLLaMA2	CLIP-L-336	Mistral-7B	54.6	51.7	51.4
VideoLLaMA2	CLIP-L-336	Mistral-8*7B	53.9	53.3	52.2
VideoChat2	UMT-L	Vicuna-7B	51.1	-	-
VideoChat2	InternVideo2 _{s3} -1B	Mistral-7B	60.3	55.8	53.0

in MVBench. In Egoschema and Perception Test evaluations, it stands out as the top performer among open-source models. When contrasted with proprietary commercial models like Gemini, VideoChat2-InternVideo2 exhibits results that are slightly lower but still comparable in the Perception Test. Egoschema requires longer context harnessing, and we plan to further explore leveraging LLM for long multimodal sequence modeling. Considering these benchmarks not only involve perception but also reasoning, these results suggest InternVideo2 does embed knowledge to partially model the world. It also validates the importance of learning a transferrable video representation for current video-related MLLM. We also give several qualitative evaluations with popular GPT-4V and GeminiPro (in supp).

6 Conclusion and Discussion

We have introduced a new family of video foundation models called InternVideo2, which achieves the state-of-the-art performance across various video and audio tasks. In InternVideo2, we combine masked video modeling, video-audio-text contrastive learning, and next token prediction into a unified framework. Additionally, we create a new video-text dataset that incorporates video-audio-speech fused captions as descriptions. The dataset contains temporally segmented clips with high semantic coherence. These designs in InternVideo2 contribute to enhancing video understanding in both perception and reasoning tasks. Notably, InternVideo2 excels in video-related dialogue and long video understanding, demonstrating its effectiveness in capturing high-level semantics.

Limitations and Discussions. Despite its achievements, InternVideo2 does not introduce specific novel architectural design. Instead, it leverages the existing learning techniques for scaling video foundation models while focusing on improving data processing to enhance its spatiotemporal perception, semantic alignment, and basic knowledge embedding. Similarly to previous studies [51, 101], InternVideo2 still grapples with limitations stemming from fixed input resolutions, sampling rates, and highly compressed tokens, which restrict its ability to express rich video information and capture fine-grained details.

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